

GAME DESIGN DOCUMENT

THE WEAVER

Pan African Game Jam · GameCloud

A dimensional platformer where a lone Weaver crosses collapsing dimensions to find the source of the Great Tear — and choose the fate of the multiverse.

Team	GameCloud
Engine	Godot 4
Platform	Mobile — iOS / Android (Portrait)
Genre	2D Platformer · Environmental Puzzle · Exploration
Art Style	Illustration / Cartoon — African-Inspired Palette
Jam Theme	Play Without Borders
Prototype Scope	1 complete dimension — L'Inversé
Run Duration	15–20 min first run · 4–5 min speedrun

Credits / Team



◆ TEAM COMPOSITION

Role	Responsibilities
Game Designer	GDD · level design · mechanic specification · vision
Dev 1	Godot physics: gravity inversion, checkpoints, CharacterBody2D
Dev 2	Needle mechanic (swipe input) · death/respawn · UI HUD
AI / ML Expert	Unstable zone expansion AI · adaptive difficulty system
Full Stack	Build pipeline · itch.io submission · final choice screen · iOS+Android export

Game Analysis

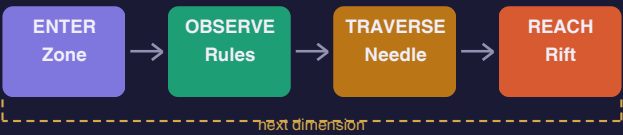


◆ STORY PREMISE

Long ago, all dimensions existed in separation. Then someone — or something — pulled on the thread. The fabric between worlds unraveled. Fragments of dimensions began to touch, merge, bleed into each other.

The Weaver belongs to the space between dimensions. Armed with the Needle, they travel through chaos — not to conquer, but to understand. Who caused the Great Tear? What should be done about it?

◆ CORE GAMEPLAY LOOP



- **Enter Zone** — Arrive in a new dimension via a tear. Physics rules immediately visible.
- **Observe Rules** — No tutorial text. Learn through exploration and failure.
- **Traverse with Needle** — Use the fold mechanic to navigate obstacles and puzzles.
- **Reach the Rift** — Cross the exit tear. Unlock next dimension.

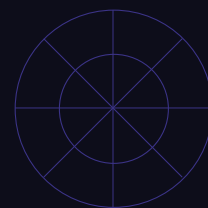
Ref: Hollow Knight (env. learning) · Celeste (die-learn-retry) · Journey (transition as event)

◆ THEME — PLAY WITHOUT BORDERS

Interpretation	In-Game Implementation
Literal: crossing dimension borders	Each zone = crossing a physical border with new physics rules
Mechanical: borders as game system	The Needle manipulates dimensional seams — borders ARE the mechanic
Cultural: no cultural borders	African-inspired aesthetics (kente, bogolan, Sahelian) in a sci-fi universe
Narrative: repair or tear borders	Final choice: mend the borders or destroy them entirely

◆ DESIGN QUESTIONS

Question	Answer
Artistic genre	Oniric · Sci-fi · African-inspired fantasy · Dark contemplative
Win condition emotion	Awe and relief crossing the Rift — or dread if you tear it further
Emotions targeted	Wonder · Tension · Curiosity · Agency
Main loop duration	8–12 min / zone (jam) · 90 min total (full game)
The fun element	Mastering impossible physics · Precision Needle · Environmental storytelling
Game scenes	Intro → Main Menu → Zone (3 sub-zones) → Final Rift → Credits



Core Mechanics



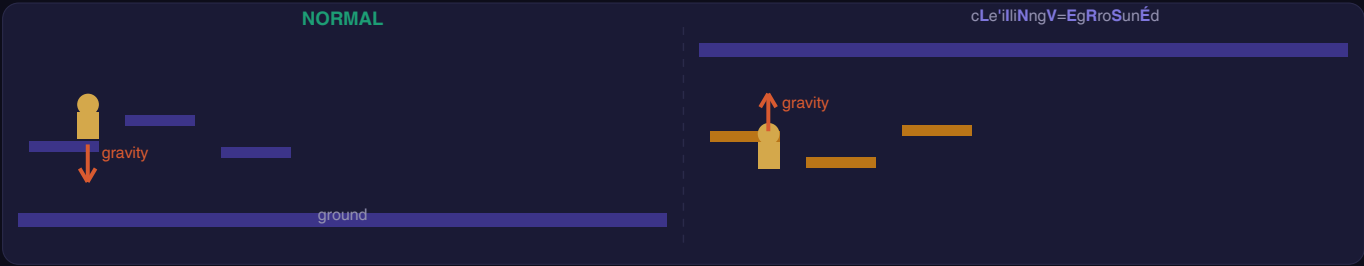
1 InvertedGravity— L'Inversé

◆ DESCRIPTION

In L'Inversé, gravity pulls upward. The player falls toward the ceiling. Jumping means diving downward. Platforms hang from the sky. No tutorial — the environment teaches by doing.

◆ RULES

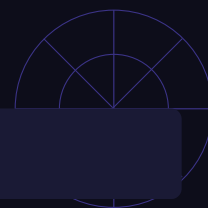
- Gravity reversed globally (**Godot: gravity_scale = -1**)
- Player **falls upward** — ceiling is the new ground
- Jump input sends player **downward** (context-inverted)
- All physics objects obey inverted gravity
- Unstable zones can appear on the floor (now above the player)



Left: normal world. Right: L'Inversé — player and objects fall toward the top of the screen.

Parameter	Value	Godot Implementation
Gravity scale	-1.0	CharacterBody2D.gravity_scale = -1.0
Jump direction	Downward	Invert jump vector Y axis
Coyote time	6 frames	Timer node — delayed on_floor check
Jump buffer	6 frames	Input buffer array, checked on landing

Ref: VVVVVV (gravity flip) · Super Mario Galaxy (local gravity)



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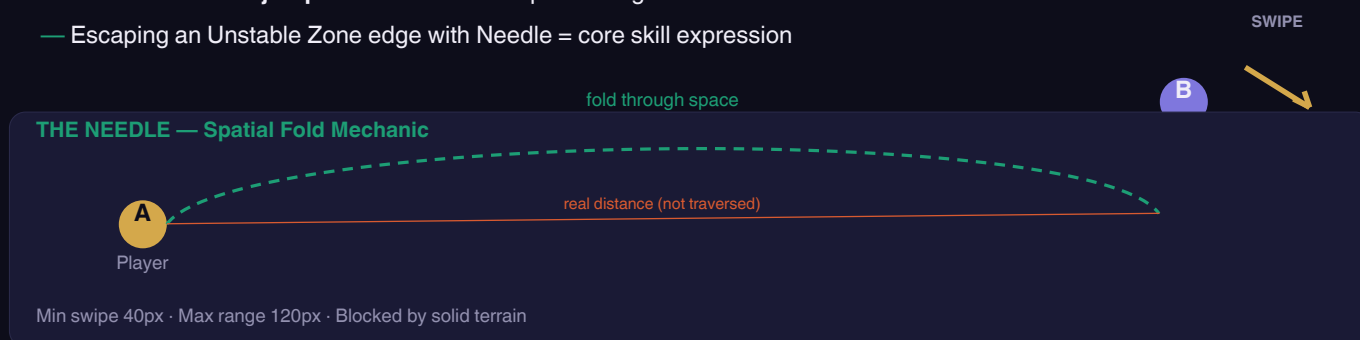
The Needle — Spatial Fold

◆ DESCRIPTION

The Weaver's primary tool. A swipe gesture creates a short-range fold in the dimensional fabric — the player is instantly displaced to a target point within range. Precise, tactical repositioning that also serves as the puzzle-solving mechanic.

◆ RULES

- Activated by **swipe gesture** — minimum threshold 40px
- Displacement range: **max 120px** from player position
- Cooldown: **2 seconds** — arc draining around the Needle icon
- Cannot fold into solid terrain — blocked fold shows rejection effect
- Can be used **mid-jump** — enables aerial repositioning
- Escaping an Unstable Zone edge with Needle = core skill expression



A → B: spatial fold (dashed arc). Red line = real distance — never traversed.

Parameter	Value	Godot Implementation
Input	Swipe (InputEventScreenDrag)	relative.length() > 40px threshold
Max range	120px	Distance check before teleport
Cooldown	2.0 seconds	Timer node + UI arc animation
Blocked feedback	Shake + glow pulse	AnimationPlayer on Needle icon
Mid-air use	Allowed	No floor check required

Ref: Celeste (dash as puzzle tool) · Portal (micro-scale) · Hyper Light Drifter

Core Mechanics (continued)



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Unstable Zones — Dynamic Threat

DESCRIPTION

Zones of decomposing dimensional fabric — visually red and vibrating. Contact = instant death. They expand slowly, creating urgency without a visible timer. The AI expert manages expansion speed, terrain pathfinding, and adaptive difficulty.

RULES

- Visual: pulsing red/crimson patches with particle edge distortion
- Contact = **instant death** → respawn at last checkpoint
- Default expansion: **2px per second**
- Expansion **blocked by platforms** — flows around terrain
- Player dies **3x at same checkpoint** → expansion drops to 1px/sec (AI adaptive)
- Two types: **static** (environmental hazard) and **dynamic** (active threat)



Parameter	Value	Godot Implementation
Shape	Polygon2D + Area2D	CollisionPolygon2D for contact
Expansion	2px/sec default	Tween on polygon vertices
AI adaptation	After 3 deaths: rate ×0.5	death_counter var per checkpoint
Terrain blocking	Yes	Raycast before vertex expansion



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Checkpoint System — Couture Seams

◆ DESCRIPTION

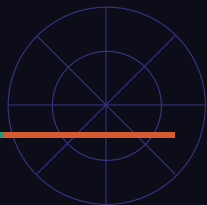
Checkpoints are visual narrative elements — glowing couture stitches in the dimensional fabric. Touching one saves progress, plays a soft audio cue, and briefly stabilizes nearby Unstable Zones.

◆ RULES

- Triggered by **Area2D overlap** — no button press needed
- Saves: player position, zone state, collected lore fragments
- Visual: stitching glow animation (1 second on activation)
- Nearby Unstable Zones **pause expansion 3 seconds** on activation
- **3 checkpoints** total in the prototype zone

Checkpoint	Location	Sub-zone unlocked	Special
CP-1	End of sub-zone 1	Sub-zone 2	Tutorial complete moment
CP-2	Mid sub-zone 2	—	Lore fragment triggers here
CP-3	Start of sub-zone 3	Final Rift approach	Zone visually destabilizes

Core Mechanics (continued)



MOBILE CONTROLS — PORTRAIT LAYOUT

Action	Input	Technical Note	Priority
Move left/right	Virtual joystick — left thumb	Dead zone 0.2	Must
Jump / Dive	Button A — right thumb	Y-inverted in L'Inversé	Must
Needle fold	Swipe in direction	40px min threshold	Must
Inspect fragment	Tap on collectible	Area2D tap detection	Nice
Pause	Tap top-right icon	Dark overlay, instant	Must

Critical mobile feel: implement **coyote time (6 frames)** and **jump buffer (6 frames)** in Godot — without these, the game will feel unjust on touchscreen and lose points on 'intuitive gameplay'.

World & Dimensions



◆ PROTOTYPE ZONE — L'INVERSÉ

Element	Description
Visual palette	Deep violet, gold, luminous white — kente & indigo inspired
Music	Reversed ambient samples, distant bass drums, no melody
Zone structure	3 sub-zones separated by checkpoint seams (glowing stitches)
Playtime	8–12 min first run · 4–5 min speedrun
Climax	The Great Rift — pulsing portal that widens. No boss combat.
Win condition	Reach the Rift and cross it (+ optional choice: repair/tear)
Fail condition	Touch Unstable Zone · Fall out of bounds

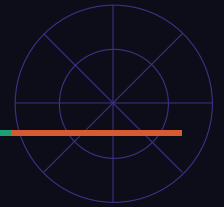
◆ ART DIRECTION — AFRICAN-INSPIRED AESTHETIC

Element	Reference & Implementation
Platforms & terrain	Kente cloth geometric patterns (Ghana) — interlocking diamonds and chevrons in gold/purple
Color palette	Bogolan/mud cloth (Mali) — deep indigo, earthy terracotta, warm ochre
Background arch.	Sahelian mosque silhouettes (Djenné, Agadez) — organic clay spires, banco walls
The Weaver	Tall slender silhouette, woven garment shifting color per dimension
The Needle	Elongated ivory/bone with geometric carvings — glows teal at use
UI / HUD	Minimal — Needle cooldown arc, health as thread unraveling, no numbers

◆ ALL SIX DIMENSIONS — FULL VISION

Dimension	Core Rule	Jam Status
L'Inversé	Gravity pulls up. Objects fall upward.	Prototype — Must
Le Muet	No sound. Thoughts appear as geometric shapes.	Bonus if time allows
Le Liquide	Floors flow. Faster you move, slower you go.	Post-jam
Le Brûlant	Colors have heat. Red burns, blue freezes.	Post-jam
L'Ancien	Time flows backward. Ruins rebuild.	Post-jam
Le Miroir	Everything mirrored. Built for multiplayer.	Post-jam (multi)

Story & Narrative



◆ STORY TREATMENT — PROTOTYPE (L'INVERSÉ)

Intro (pre-menu)

Static shot of a pulsing tear in reality. Silence. A faint sound — something being pulled. The Weaver falls upward through the screen. Cut to title.

Sub-zone 1 (learning)

Player discovers the inversion. No text. A platform that should be below is above. First Unstable Zone in the floor. CP-1 reached — soft sound, visual calm.

Sub-zone 2 (lore)

A floating shard near CP-2. The Weaver touches it. Brief vision: who lived here before the inversion. 3 seconds of imagery. No words.

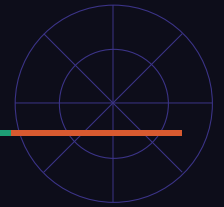
Sub-zone 3 (climax)

The Great Rift pulses and widens. Unstable Zones expand faster. The zone visually collapses. The Weaver reaches the Rift.

Final choice

Two gestures on screen — Needle inward (repair) or outward (tear further). Two visual endings. No text. Player interprets.

11-Day Development Plan



◆ SPRINT TIMELINE

Days 1–2

Setup & Movement Prototype

- Godot repo + scene structure (Main, Zone_Inversé, UI, GameManager)
- CharacterBody2D: movement + inverted gravity (gravity_scale = -1)
- Placeholder tileset — geometry only, no art yet
- Basic death zone (red Area2D trigger)
- ✓ Deliverable: player falls UP, no crash, no clip through ceiling

Days 3–4

Needle + Unstable Zones

- Swipe input: InputEventScreenDrag + 40px threshold + fold teleport
- Cooldown 2s + UI arc animation on Needle icon
- Unstable zone Polygon2D expansion via Tween (AI system)
- Death + respawn at last checkpoint
- Coyote time (6f) + jump buffer (6f)
- ✓ Deliverable: full die/respawn/needle loop working

Days 5–6

Level Design + Art Pass 1

- Paper design first: 3 sub-zones, platforms, zones, checkpoints
- Build from paper design in Godot
- Cartoon tileset: kente/indigo palette — platforms, background, terrain
- Character animations: idle, run, jump, needle-use, death
- Checkpoint glow animation (stitching effect)
- ✓ Deliverable: zone playable end-to-end

Days 7–8

Lore + Sound + Juice

- 3–5 lore fragment images (static illustrated visuals)
- Ambient music: reversed samples + bass drums
- SFX: Needle activation, checkpoint, death, zone hum
- Particle FX: zone edges, needle fold flash, checkpoint stitch
- Title screen + end screen + final choice (if time allows)
- ✓ Deliverable: complete experience from title to credits

Days 9–10

Playtests + Polish

- 5+ external playtests — NOT team members
- Log: death locations + Needle comprehension
- Tune: expansion speed, Needle range, checkpoint placement



- Critical bug fixes only — NO new features after day 8
- Mobile build tested on real device (iOS + Android)
- ✓ Record 60–90s gameplay video + portrait screenshots

Day 11

Submission

- EXE build + APK/IPA finalized and tested
- itch.io page: description + African aesthetic + double theme
- GDD submitted with all required assets
- ■ 4-hour buffer — NO code changes on submission day

Scope rule: Day 6 and zone not playable → cut lore + final choice. Day 8 and no sound → ship silent.
Non-negotiable: inverted gravity + Needle + 1 complete zone on mobile.

Scoring Strategy



◆ SCORE BREAKDOWN & STRATEGY

Criterion	Pts	Our Advantage	Risk
Game Prototype	60	Clear loop · original mechanics · native mobile controls + coyote time	Mobile calibration in 11 days
Theme Adoption	20	Double reading: literal (dimension borders) + metaphorical. 'Borders' = seams = core mechanic.	None — strongest point
Creativity	10	African aesthetic in sci-fi = globally rare. Env. lore with no text.	Ambitious art in 11 days
Commercial	10	Post-jam: 6 dimensions, multiplayer, seasonal arcs. Target: mobile 16–30.	Competitive mobile market
Bonus Cultural	+5	African team + kente/bogolan/Sahelian aesthetics integrated into gameplay.	Must be explicit in submission

◆ REFERENCES

Game	Developer	What We Take
Hollow Knight	Team Cherry	Environmental storytelling · atmospheric platformer · lore without dialogue
Celeste	Maddy Thorson	Precise mobile controls · coyote time · difficulty through environment
Manifold Garden	William Chyr	Impossible physics per zone · architecture as puzzle
Gris	Nomada Studio	Wordless narrative · color as world state · contemplative platformer
Journey	thatgamecompany	Transition as aesthetic event · silence as language
VVVVVV	Terry Cavanagh	Pure gravity inversion — direct reference for L'Inversé

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